FIREBUDDY

AN AUTOMATIC FIRE EXTINGUISHING ROBOT

A Project Case Study

PROJECT DOMAIN	Robotics • Internet of Things • Embedded Systems
PROJECT TYPE	Final-year academic project
PROJECT STATUS	Physical working prototype
OPERATING MODES	Autonomous and Wi-Fi-based manual control
CORE TECHNOLOGY	Arduino Uno • ESP32 • Embedded C/C++

AUTOMATED INITIAL FIRE RESPONSE THROUGH ROBOTICS AND IoT

1. Executive Summary

Fire accidents pose significant threats to human life, infrastructure and the environment. The time required to identify a fire, reach its location and initiate extinguishing measures can influence the extent of damage. Conventional firefighting methods may require personnel to enter hazardous environments, exposing them to extreme temperatures, smoke and other dangers.

FIREBUDDY is a physical robotic prototype that combines embedded systems, robotics and Internet of Things (IoT) technology to automate initial fire-response operations. Multiple flame sensors detect fire and help identify its direction. An Arduino Uno processes sensor inputs and controls movement, while an ultrasonic sensor assists with obstacle detection. An L298N motor driver operates the DC motors.

The robot supports autonomous mode and Wi-Fi-based manual mode. In autonomous mode, it detects flames, navigates towards the detected fire and activates its extinguishing mechanism at a suitable position. In manual mode, an operator can remotely control the robot using the ESP32 wireless functionality.

A relay module controls the water pump, and a servo motor adjusts the water-spraying direction. The flame sensors continue monitoring during extinguishing. When the sensors indicate that the fire is no longer detected, the controller deactivates the pump.

2. Problem Statement

Fire emergencies require immediate detection and an effective response. Conventional approaches face several challenges:

- **Risk to human life:** Firefighters may enter environments containing extreme heat, smoke and hazardous substances.
- **Delayed initial response:** Manual detection and intervention can introduce delays, allowing a fire to spread.
- **Difficult-to-access environments:** Narrow pathways, obstacles and unstable surroundings can restrict access.
- **Dependence on human intervention:** Traditional response methods often require personnel to remain near the affected location.
- **Need for automation:** An automated system could detect flames, approach the affected area and initiate an appropriate response with reduced direct human involvement.

Problem addressed: The need for an automated robotic prototype that can detect fire, navigate through its surroundings and initiate water-based extinguishing with minimal human intervention.

3. Proposed Solution

FIREBUDDY integrates fire-detection sensors, navigation components, an embedded controller and an automatic extinguishing mechanism. Multiple flame sensors monitor different directions. The Arduino Uno compares their readings to identify the approximate direction of the detected fire and coordinates movement.

An ultrasonic sensor monitors nearby obstacles. The L298N motor driver controls the DC motors. When the robot reaches a suitable position, a relay activates the water pump and a servo motor adjusts the nozzle direction. The flame sensors continue monitoring during extinguishing, and the pump is switched off when fire is no longer detected.

The ESP32 provides Wi-Fi connectivity for remote manual operation.

Core Capabilities

- Multiple-sensor flame detection and approximate fire-direction identification.
- Autonomous robotic navigation and ultrasonic obstacle detection.

- Automatic water-pump activation and flame-based deactivation.
- Servo-controlled water-spray direction.
- Wi-Fi-based remote manual control and dual operating modes.

4. Project Objectives

- Develop a physical robotic prototype for automated fire detection and extinguishing.
- Use multiple flame sensors to detect fire and identify its approximate direction.
- Implement autonomous movement based on sensor inputs.
- Integrate ultrasonic sensing for obstacle detection and avoidance.
- Automate water-pump activation and deactivation.
- Implement servo-based directional water spraying.
- Integrate ESP32 for Wi-Fi-based remote operation.
- Provide autonomous and manual operating modes.
- Reduce the need for direct human involvement during initial fire-response operations.

5. Hardware Components and Functions

Component	Role in FIREBUDDY
Arduino Uno	Primary controller and decision-making unit.
ESP32	Wi-Fi communication and remote-control functionality.
Multiple flame sensors	Fire detection and approximate direction identification.
Ultrasonic sensor	Obstacle detection and distance measurement.
L298N motor driver	Controls the DC motors.
DC motors	Provide robotic movement.
Relay module	Electrical switching for the water pump.
Water pump	Delivers water through the extinguishing nozzle.
Servo motor	Adjusts the direction of the water spray.
Robot chassis	Supports and houses the hardware.
Power supply	Provides electrical power to the system.

5.1 Arduino Uno – Primary Processing Unit

The Arduino Uno acts as the main controller. It receives information from the flame and ultrasonic sensors, processes readings and executes programmed instructions. Its responsibilities include flame-signal processing, approximate fire-direction identification, navigation coordination, obstacle-distance processing, motor control, pump switching and servo positioning.

5.2 ESP32 – Wireless Communication

The ESP32 provides Wi-Fi connectivity for remote manual operation. It allows an operator to send movement commands without physically interacting with the robot. The exact control interface and communication path depend on the implemented hardware and software.

5.3 Multiple Flame Sensors – Fire Detection

Multiple flame sensors monitor different directions around the robot. Each detects infrared radiation associated with flames and provides a corresponding signal. By comparing readings from the sensors, the controller estimates the direction of the detected fire and adjusts movement accordingly.

5.4 Ultrasonic Sensor – Obstacle Detection

The ultrasonic sensor measures the distance to nearby objects by transmitting ultrasonic waves and measuring the time taken for reflected waves to return. The controller uses this information to decide whether the robot can continue moving or needs to execute an avoidance movement.

5.5 L298N Motor Driver – Movement Control

The L298N acts as an interface between the Arduino Uno and DC motors. It receives control signals and enables programmed movements such as forward, reverse, left, right and stop.

5.6 Relay Module and Water Pump – Automatic Extinguishing

The relay acts as an electrically controlled switch for the water pump. When the robot reaches a suitable extinguishing position, the Arduino activates the relay and the pump delivers water through the nozzle. The pump is deactivated when the flame-sensor logic indicates that fire is no longer detected.

5.7 Servo Motor – Directional Water Spraying

The servo motor adjusts the orientation of the water-spraying mechanism, helping direct water towards the detected fire rather than spraying in only one fixed direction.

6. System Architecture

FIREBUDDY consists of five interconnected functional modules:

- **Fire detection:** Multiple flame sensors provide readings to the Arduino Uno, which estimates the presence and approximate direction of flames.
- **Navigation and obstacle avoidance:** The ultrasonic sensor monitors nearby obstacles and supports avoidance decisions.
- **Movement control:** The Arduino sends signals to the L298N motor driver, which operates the DC motors.
- **Fire extinguishing:** The relay activates the water pump, while the servo adjusts spray direction. Flame monitoring continues during operation.
- **Wireless communication:** The ESP32 provides Wi-Fi connectivity for remote manual control.

Simplified System Flow

AUTONOMOUS NAVIGATION	FIRE EXTINGUISHING	MANUAL CONTROL
Flame sensors + ultrasonic sensor ↓ Arduino Uno ↓ Direction and obstacle decisions ↓ L298N motor driver ↓ DC motors and navigation	Fire detected ↓ Reach suitable position ↓ Relay ON ↓ Water pump + servo ↓ Monitor flame sensors ↓ No flame detected: pump OFF	Mobile/Wi-Fi interface ↓ ESP32 ↓ Control commands ↓ Robot operation

7. Working Methodology

Step 1 – System initialization

When powered on, the Arduino initializes the connected sensors, motor driver, relay and servo. The ESP32 establishes its Wi-Fi communication functionality.

Step 2 – Continuous fire monitoring

Multiple flame sensors monitor their respective detection areas and provide readings to the controller.

Step 3 – Fire-direction identification

The Arduino compares sensor readings to estimate the approximate direction of the detected flame, based on sensor arrangement and calibration.

Step 4 – Autonomous navigation

The robot moves towards the detected fire. The Arduino sends movement instructions to the L298N motor driver, which controls the DC motors.

Step 5 – Obstacle detection and avoidance

The ultrasonic sensor measures nearby distances. If an obstacle is within the programmed threshold, the robot executes its programmed avoidance movement.

Step 6 – Reaching the fire

The robot approaches a suitable position for extinguishing, maintaining an appropriate distance for safe operation.

Step 7 – Automatic pump activation

The Arduino activates the relay, switching on the water pump and delivering water through the nozzle.

Step 8 – Directional water spraying

The servo motor adjusts the nozzle orientation to direct water towards the detected fire.

Step 9 – Continuous fire monitoring

The flame sensors continue monitoring while extinguishing is in progress.

Step 10 – Automatic pump deactivation

When the flame-sensor logic indicates that fire is no longer detected, the Arduino deactivates the relay and switches off the water pump.

8. Operating Modes

8.1 Autonomous Mode

In autonomous mode, the robot performs its programmed operations without requiring continuous manual movement commands.

- Initialize the system and monitor the surroundings.
- Detect flames and estimate their direction.
- Navigate towards the detected fire.
- Detect and avoid obstacles.
- Reach a suitable position and activate the water pump.
- Adjust spray direction using the servo motor.

- Monitor the flame sensors and stop the pump when fire is no longer detected.

8.2 Wi-Fi-Based Manual Mode

In manual mode, the robot is controlled remotely through the ESP32 Wi-Fi functionality. An operator sends movement commands through the implemented control interface. This mode is useful when autonomous navigation is unsuitable, when the operator needs to position the robot manually, or when human intervention is required.

9. Software Implementation

9.1 Arduino Uno Programming

The Arduino executes the primary robotic control logic, including hardware initialization, reading multiple flame sensors, comparing readings, estimating fire direction, reading ultrasonic distance measurements, obstacle avoidance, motor control, relay activation and deactivation, flame monitoring and servo positioning.

9.2 ESP32 Programming

The ESP32 provides Wi-Fi connectivity for remote manual control. It establishes wireless communication, receives movement commands and supports remote robot operation through the implemented interface.

9.3 Embedded Systems Integration

The project coordinates sensor inputs, movement outputs and the extinguishing mechanism through an embedded control architecture. The Arduino Uno acts as the primary controller, while the ESP32 provides wireless connectivity. The precise communication interface between the two controllers should be documented according to the actual wiring and program.

10. Testing and Evaluation

The physical prototype should be evaluated through individual component tests and integrated system tests. The following table describes the test plan and expected behaviour.

Test case	Testing objective	Expected behaviour
Multiple flame sensors	Verify detection from different directions.	Detect the test flame.
Fire-direction test	Compare sensor readings.	Estimate approximate flame direction.
Ultrasonic sensor	Measure obstacle distance.	Detect nearby objects.
Motor control	Test directional movement.	Perform programmed movements.
Autonomous navigation	Test movement towards fire.	Navigate according to sensor readings.
Obstacle avoidance	Introduce obstacles.	Execute avoidance movements.
Relay and pump	Verify switching and water delivery.	Activate and deactivate water flow.
Servo motor	Check nozzle positioning.	Adjust spray direction.
Automatic pump stop	Remove test flame during operation.	Stop pump when no flame is detected.
Wi-Fi control	Send remote movement commands.	Respond to operator commands.
Integrated prototype	Test complete operating sequence.	Perform the programmed response workflow.

Performance Evaluation Metrics

Fire Detection Accuracy

Fire Detection Accuracy (%) = (Successful Detection Trials ÷ Total Detection Trials) × 100

Fire Response Time
Time between initial fire detection and activation of the extinguishing mechanism.

Response Time Reduction
((Baseline Response Time – Robot Response Time) ÷ Baseline Response Time) × 100

Obstacle Avoidance Success Rate
Percentage of obstacle-avoidance trials completed successfully.

Automatic Pump Deactivation Time
Time between the fire no longer being detected and the water pump switching off.

Wi-Fi Control Reliability
Percentage of remote-control commands successfully received and executed.

Actual performance values should be obtained from repeated, documented experiments rather than estimated.

11. Results and Observations

FIREBUDDY was developed as a physical working prototype integrating multiple sensors, embedded controllers, motor-control components and an automated extinguishing mechanism. Its implemented design includes:

- Multiple flame sensors for detecting flames from different directions.
- Sensor-based estimation of the approximate fire direction.
- Autonomous robotic navigation and ultrasonic obstacle detection.
- DC motor control through the L298N motor driver.
- Automatic water-pump activation through a relay.
- Servo-controlled directional water spraying.
- Automatic pump deactivation when flame sensors no longer detect fire.
- Wi-Fi-based remote manual control through ESP32.
- Autonomous and manual operating modes.

The prototype demonstrates integration of these capabilities into a single robotic system. Quantitative results such as detection accuracy, response-time reduction and obstacle-avoidance success rate should be reported only after documented testing.

12. Technical Challenges and Considerations

12.1 Multiple Sensor Coordination

The controller must interpret readings from multiple flame sensors. Sensor calibration, positioning and reliable signal comparison are important for estimating fire direction.

12.2 Navigation Accuracy

The robot's approach depends on correct interpretation of sensor readings and turning accuracy. Movement logic should account for changes in sensor readings.

12.3 Obstacle Avoidance

The robot must respond appropriately when an obstacle is detected. Distance thresholds and motor responses should be tested under different conditions.

12.4 Pump Control

The extinguishing mechanism must activate when required and deactivate when fire is no longer detected. The control logic should account for temporary sensor fluctuations.

12.5 Wireless Communication

Manual operation depends on reliable communication between the control interface and ESP32. Connectivity, response time and command reliability should be tested.

12.6 Hardware Integration

The sensors, controllers, motors and pump must operate together. Proper wiring, power distribution and component protection are essential.

13. Safety Considerations and Limitations

FIREBUDDY is an academic working prototype intended to demonstrate automated fire detection and water-based extinguishing. It is not a certified firefighting device and should not replace professional firefighting equipment.

- Flame sensors have limited detection ranges and can be affected by environmental lighting.
- Sensor-based direction identification may not work reliably in every environment.
- Ultrasonic sensors cannot guarantee collision-free navigation in all conditions.
- Wi-Fi-based manual control depends on wireless connectivity.
- The system does not use a camera or computer-vision-based fire classification.
- Flame detection alone does not independently confirm that a fire is completely extinguished.
- Water-based extinguishing is unsuitable for certain fire classes.

Important safety warning: Water must not be sprayed onto energized electrical equipment or flammable-liquid fires where it could spread the fire. Further development should include fire-class suitability, independent safety mechanisms, emergency stopping, thermal protection and an extinguishing agent appropriate for the intended application.

14. Future Enhancements

14.1 AI-Based Fire Detection

Machine learning and computer vision could be integrated to improve fire classification and reduce false-positive detections.

14.2 Thermal Imaging

A thermal camera could help identify heat sources and improve detection in environments with limited visibility.

14.3 Smoke and Gas Detection

Additional smoke and gas sensors could provide more information about the surrounding environment.

14.4 Real-Time Monitoring Dashboard

A web or mobile dashboard could display robot status, fire alerts, sensor readings, battery status, movement status, Wi-Fi connectivity and extinguishing-system status.

14.5 Advanced Autonomous Navigation

Advanced navigation algorithms could improve movement and obstacle avoidance in complex environments.

14.6 Improved Extinguishing Mechanism

Future versions could incorporate an extinguishing mechanism selected according to fire classification and environmental requirements.

14.7 Enhanced Communication

Additional communication methods could improve remote operation and monitoring where Wi-Fi connectivity is unreliable.

15. Project Impact

- **Reduced human exposure:** Automated initial response can reduce the need for a person to approach the fire directly during every stage.
- **Automated operation:** Multiple flame sensors, obstacle detection and automatic pump control enable a programmed response sequence.
- **Flexible control:** Autonomous and Wi-Fi-based manual modes provide operational flexibility.
- **Practical engineering integration:** The prototype combines sensors, embedded programming, motor control, wireless communication and mechanical components.
- **Future scalability:** The architecture can be extended with AI-based detection, thermal imaging and advanced navigation.

16. Conclusion

FIREBUDDY – An Automatic Fire Extinguishing Robot – demonstrates how robotics, embedded systems and Internet of Things technology can be integrated to develop an automated initial fire-response system.

The physical working prototype combines multiple flame sensors for fire detection and directional identification, an Arduino Uno for primary control, an ultrasonic sensor for obstacle detection and an ESP32 for Wi-Fi-based manual operation.

Its automated extinguishing mechanism uses a relay-controlled water pump and a servo motor to direct water towards the detected fire. The system monitors the flame sensors during extinguishing and deactivates the pump when fire is no longer detected.

By supporting autonomous and manual operating modes, FIREBUDDY provides flexibility while demonstrating practical hardware and software integration. Although further testing, safety mechanisms and performance validation are required before real-world deployment, the project establishes a foundation for further exploration of intelligent robotic systems for emergency response.

FIREBUDDY represents an effort to make initial fire-response operations more automated and safer by using technology to reduce direct human exposure to hazardous environments.

17. Project Information Summary

Category	Details
Project name	FIREBUDDY
Project title	An Automatic Fire Extinguishing Robot
Project type	Physical working prototype
Domain	Robotics, IoT and Embedded Systems
Main controller	Arduino Uno
Wireless module	ESP32
Programming	Embedded C/C++
Fire detection	Multiple flame sensors
Fire direction identification	Comparison of multiple flame-sensor readings
Obstacle detection	Ultrasonic sensor
Motor driver	L298N
Extinguishing mechanism	Relay-controlled water pump
Spray direction	Servo motor
Operating modes	Autonomous and Wi-Fi-based manual control
Camera	Not implemented
Primary application	Automated initial fire response

Keywords: FIREBUDDY, Fire Extinguishing Robot, Arduino Uno, ESP32, Internet of Things, Multiple Flame Sensors, Autonomous Navigation, Obstacle Avoidance, Embedded Systems, Robotics, Automated Fire Detection.